

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 11



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Recap of Previous Lecture



- ✓ Topic Complemented lattice
- ✓ Topic Sub-lattice
- ✓ Topic Distributive lattice
- ✓ Topic Boolean algebra

Topics to be Covered



✓ **Topic**

Functions

✓ **Topic**

Total number of functions

✓ **Topic**

One-one and onto functions

✓ **Topic**

Number of one-one functions

✓ **Topic**

Number of onto functions



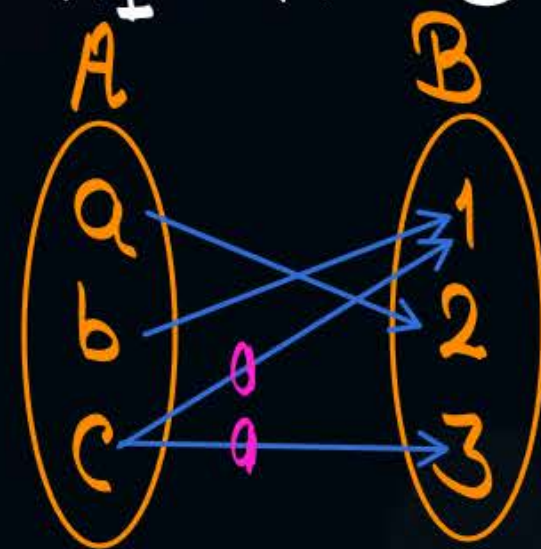
Topic : Function

Every function from set A to set B is also a relation from A to B, but Every relⁿ from A to B need not be a function from set A to set B



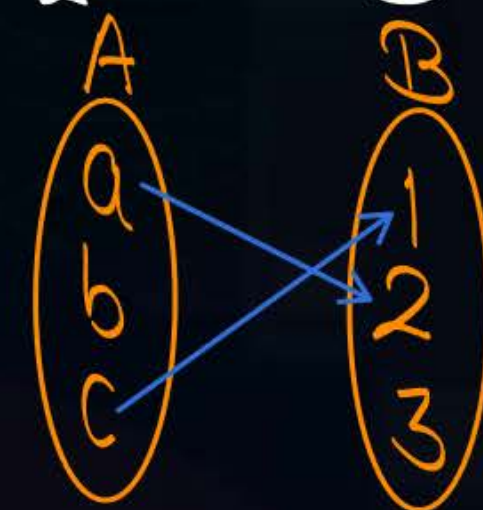
A relation from set A to set B is called a function from set A to set B if and only if Every element of set A relates with exactly '1' element of set B.

eg: $R_1: A \rightarrow B$ { R_1 is a relation }
from A to B

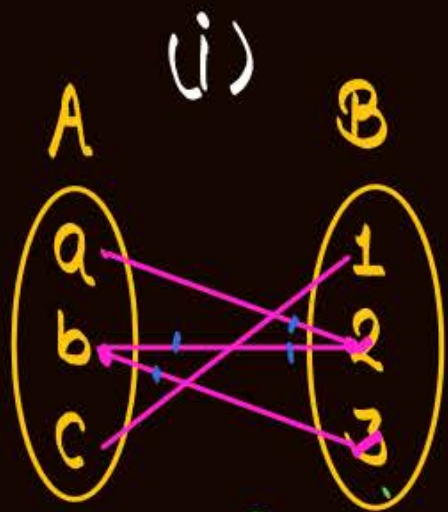


{ element 'c' of set A relates with two elements of set B.
∴ Not a function

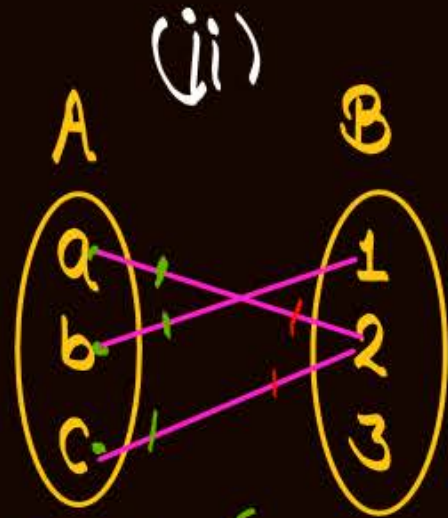
eg: $R_2: A \rightarrow B$ { R_2 is a relation }
from A to B



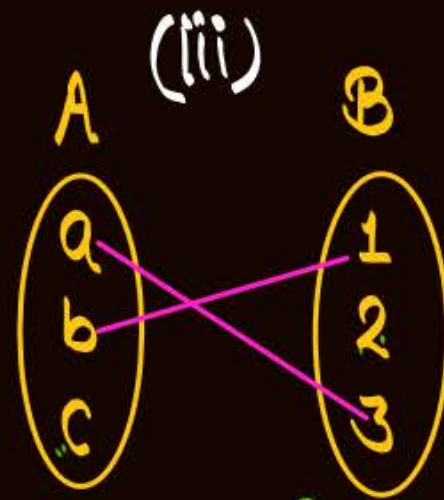
{ Element 'b' relates with '0' element of set B.
∴ Not a function



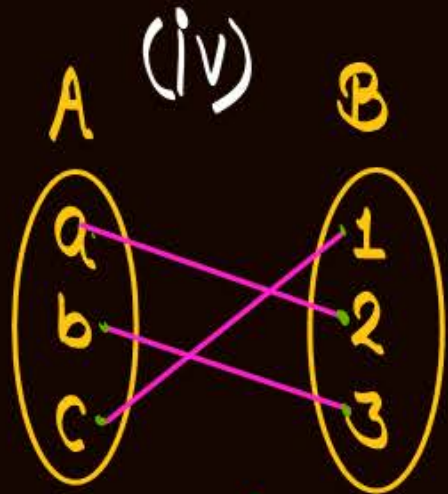
Neither a function
from set A to B
nor a function from B to A



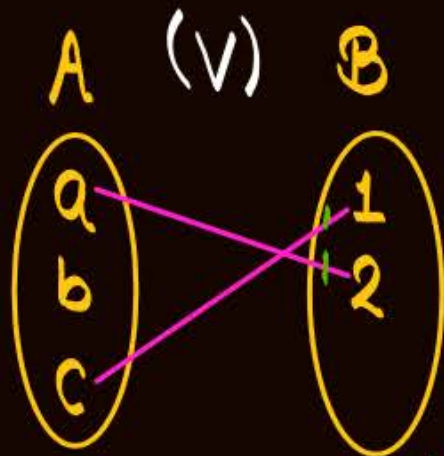
It is function
from A to B
but not a function
from B to A



Neither a function
from set A to B
nor a function from B to A



It is function
from A to B
as well as a function
from B to A



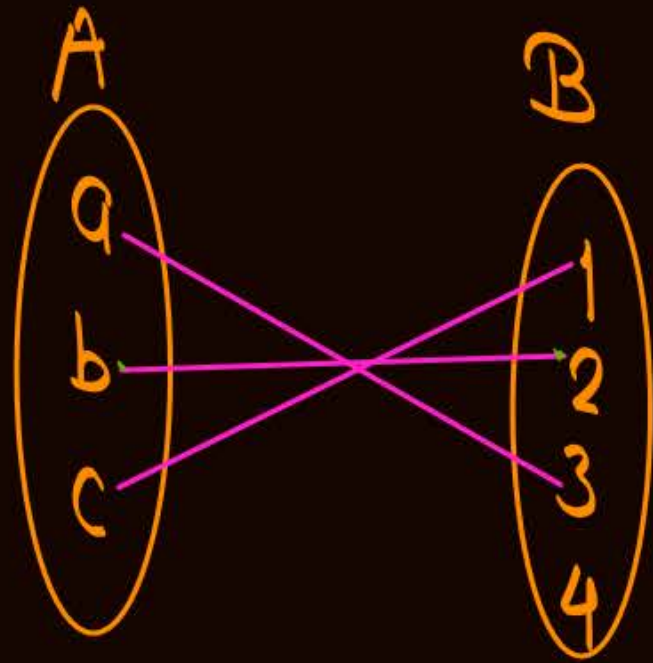
It is not a function
from A to B
but it is a function from B to A.

→ Let $f: A \rightarrow B$ is a function,

then set 'A' is called domain of the function 'f'

and set 'B' is called Co-domain of the function 'f'

eg. $f: A \rightarrow B$



Domain of $f = \{a, b, c\}$

Co-domain of $f = \{1, 2, 3, 4\}$

$f(b) = 2$,

ie, image of element 'b' w.r.t. function 'f' is '2'.

Or, Pre-image of element '2' w.r.t. function 'f' is 'b'.



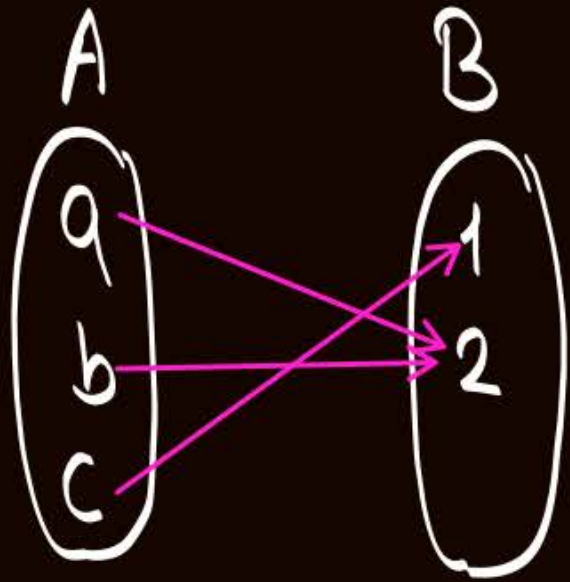
Topic : Range of a function

→ The set of all elements from Co-domain that are mapped by at least one element of domain is called "Range" of the function.

$$\text{Range of function} \subseteq \text{Co-domain of function}$$

eg:

$$f_1: A \rightarrow B$$



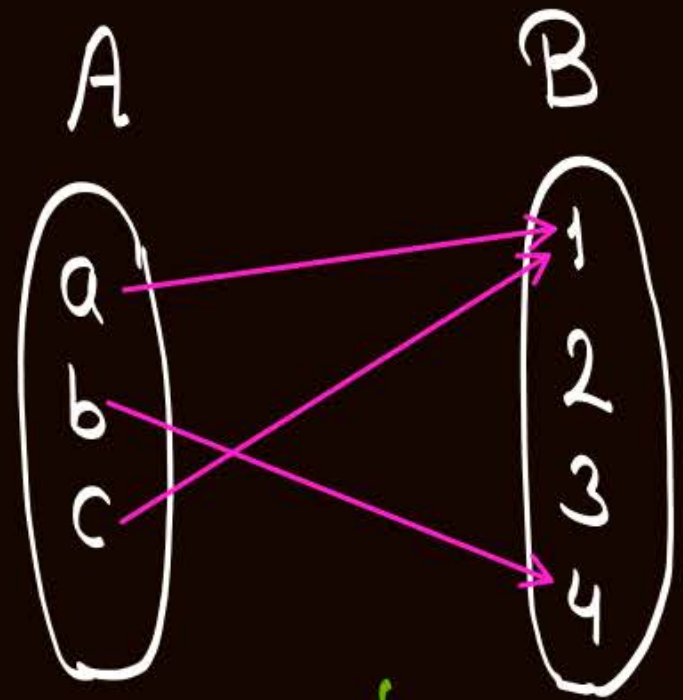
Domain: $\{a, b, c\}$

Co-domain: $\{1, 2\}$

Range = $\{1, 2\}$

Range = Co-domain

$$f_2: A \rightarrow B$$



Domain: $\{a, b, c\}$

Co-domain = $\{1, 2, 3, 4\}$

Range = $\{1, 4\}$

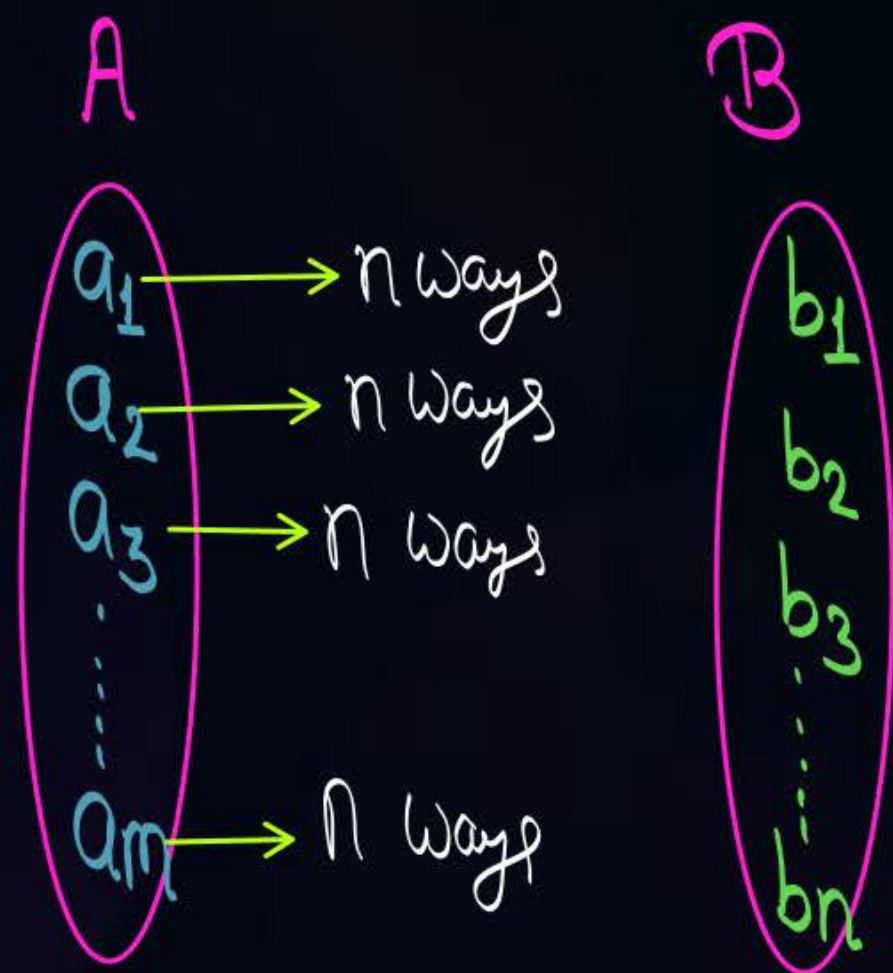
Range $\subset$ Co-domain



Topic : Total number of functions

* Let $|A|=m$ and $|B|=n$, then

Total Number of functions Possible from set A to set B = $n^m = |B|^{|A|}$
(Size of co-domain)
(Size of domain)



Total no. of functions from A to B

$$= \underbrace{n \times n \times n \times \dots \times n}_{\text{'m' times 'n'}} = n^m$$

Note:-

A function defined from set A to set A itself is called a function on set A .

If $|A| = n$, then

Total no. of functions possible on set A _{from A to A } $= |A|^{|A|} = n^n$

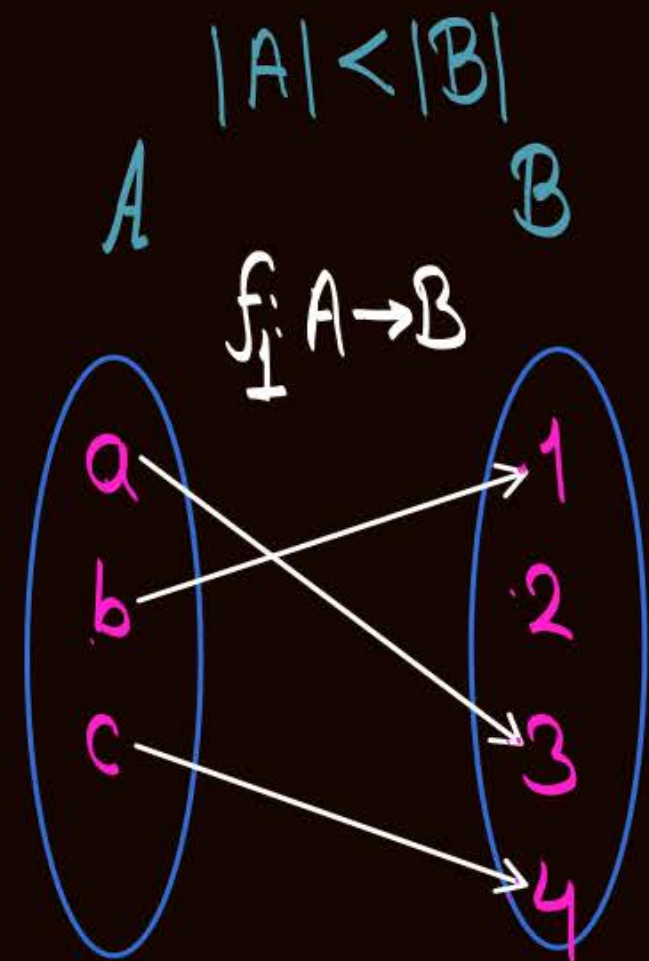


Topic : Injective (one-one) function

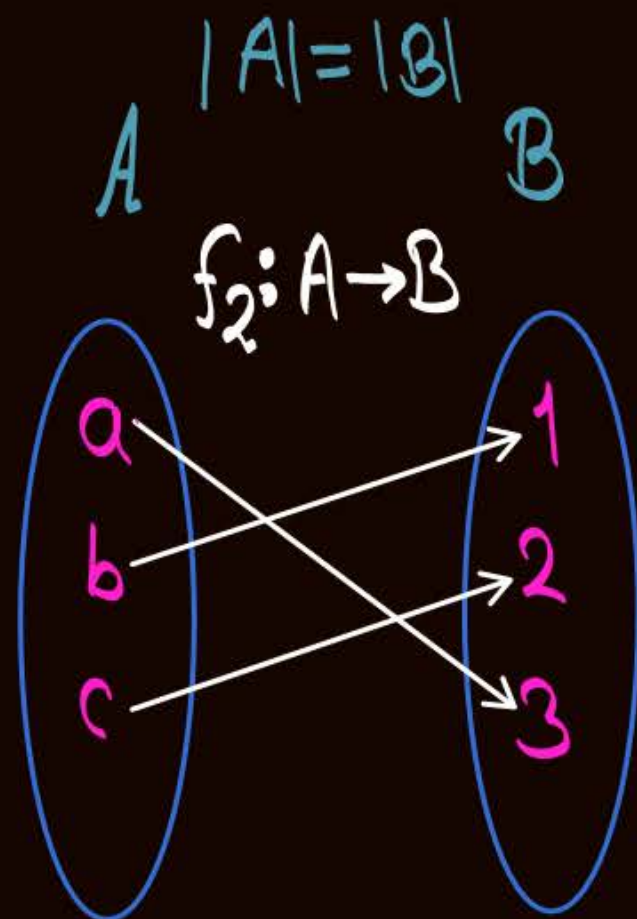
A function $f: A \rightarrow B$ is called an injective (one-one) function if and only if.

distinct elements of the domain (set A) have distinct images in Codomain (set B)

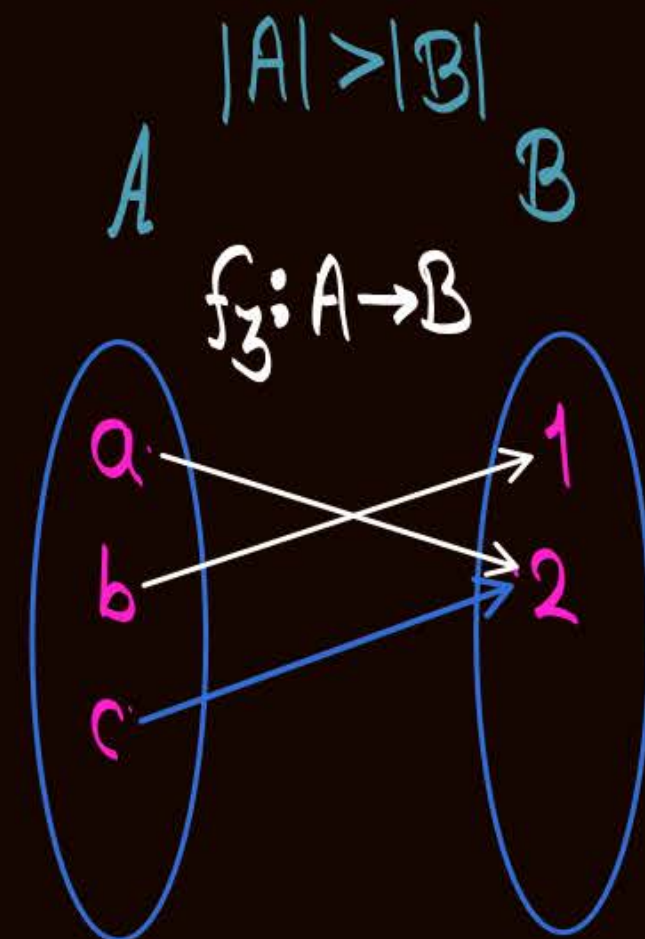
i.e. $\forall a, b \in \text{Domain}$
if $a \neq b$ then $f(a) \neq f(b)$



' f_1 ' is a
One-one function



' f_2 ' is a
One-one function



' f_3 ' is not a
One-one function
 $a \neq c$ but $f(a) = f(c)$

Note:-

① If $|A| > |B|$, then
One-one function is not possible from set A to set B

② If One-one function exists from set A
to set B, then $|A| \leq |B|$

③ In one-one function Every element of
Co-domain can have at most '1' pre-image

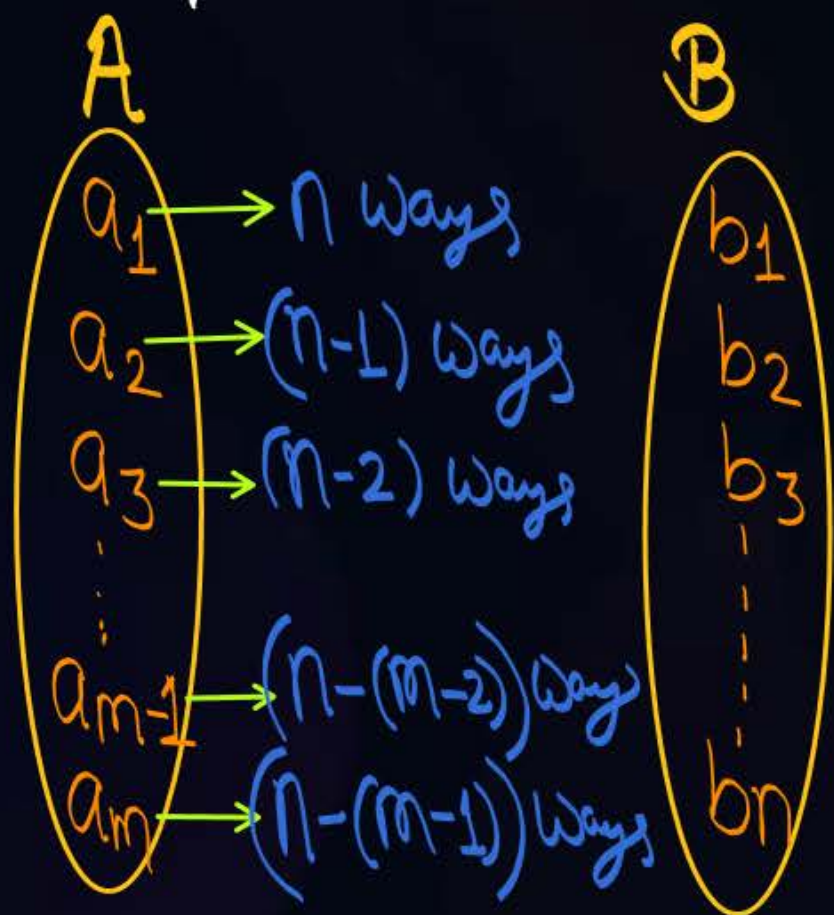


Topic : Number of one-one function

Let $|A| = m$ & $|B| = n$ (such that $m \leq n$),
then,

$${}_nP_r = \frac{n!}{(n-r)!}$$

Number of one-one functions possible from set A to set B = $\frac{n!}{(n-m)!} = {}_nP_m$



Total no. of one-one function possible from set A to set B = $n(n-1)(n-2)\dots(n-m+1)$

$$= n(n-1)(n-2)\dots(n-m+1) \times \frac{(n-m)(n-m-1)\dots 3 \cdot 2 \cdot 1}{(n-m)(n-m-1)\dots 3 \cdot 2 \cdot 1}$$
$$= \frac{n!}{(n-m)!}$$

$${}_nC_m \times m! = {}_nP_m$$

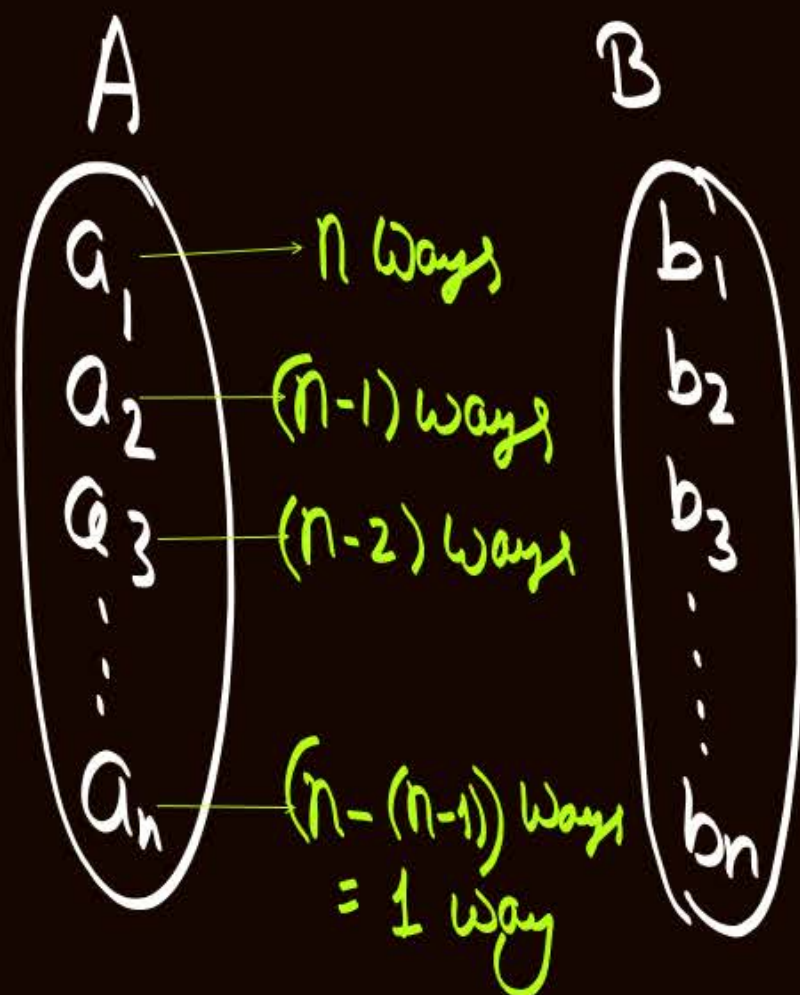
because ${}_nC_r = \frac{n!}{r!(n-r)!}$

Note:

If $|A| = |B| = n$,

$$0! = 1$$

then, number of one-one function
Possible from set A to set B $= {}^n P_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$



Total no. of
one-one fn
possible

$$= n(n-1)(n-2) \dots 1$$
$$= n!$$



Topic : Surjective (onto) function

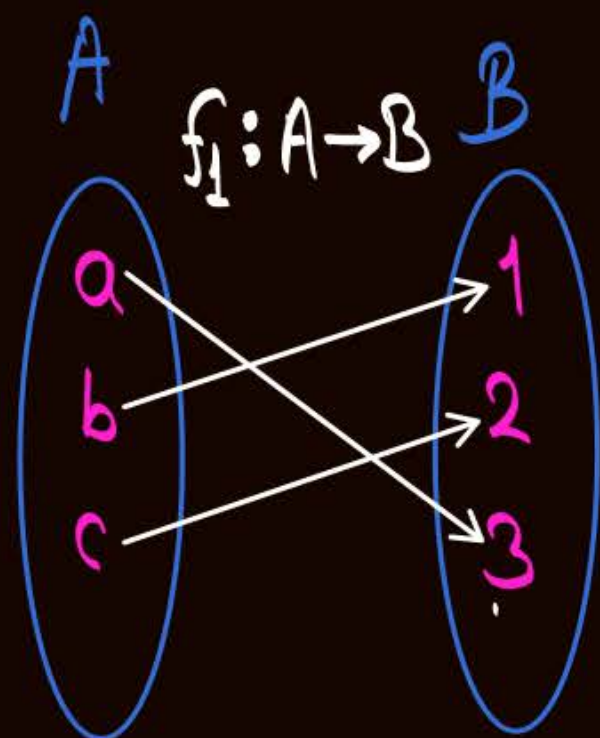
A function $f: A \rightarrow B$ is called an Onto (Surjective) function if and only if

Every element of Co-domain is mapped by at least one element of domain

ie, In an onto function

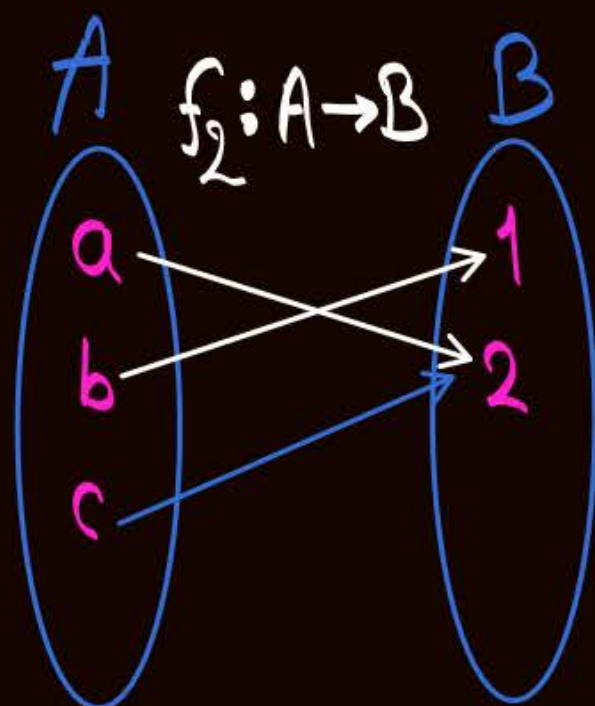
Range of function = Co-domain of function

$$|A| = |B|$$



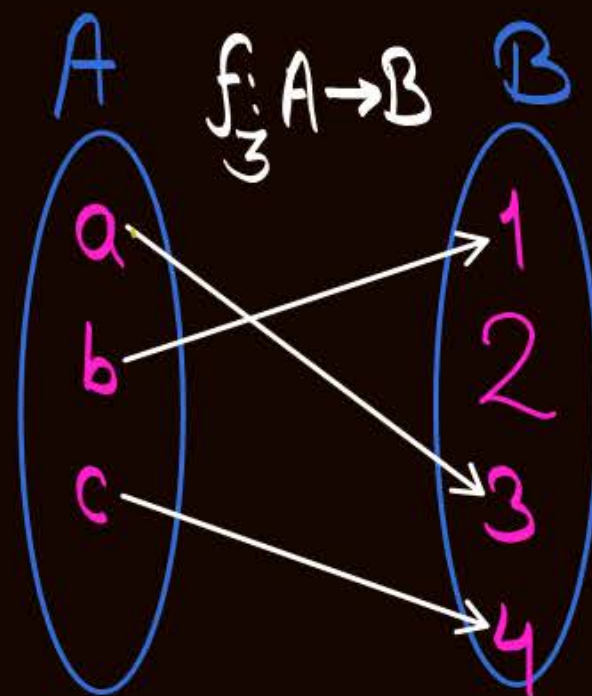
Range = Co-domain
 $\therefore f_1$ is an
 Onto function

$$|A| > |B|$$



Range = Co-domain
 $\therefore f_2$ is an
 Onto function

$$|A| < |B|$$



Range $\subset$ Co-domain
 $\therefore f_3$ is not an
 Onto function

Note:

- ① If $|A| < |B|$, then
Onto function is not possible from set A to set B.
- ② If there exist an onto function from
set A to set B, then $|A| \geq |B|$
- ③ In an onto function Every element of
Co-domain will have at least '1' pre-image



Topic : Number of onto functions

① If $|A| = |B| = n$, then

How many Onto functions are possible from A to B

② If $|A| = n$ & $|B| = 2$ such that $n \geq 2$, then

How many Onto functions are possible from A to B

③ if $|A| = 4$ & $|B| = 3$, then

How many onto functions are possible from A to B.

Q:- If $|A| = |B| = n$
then How many onto functions are possible from A to B.

Note:- If $|A| = |B|$, then
Every one-one function from set A to set B is onto as well.
and Every onto function from set A to set B will be one-one as well.

∴ When $|A| = |B|$, then

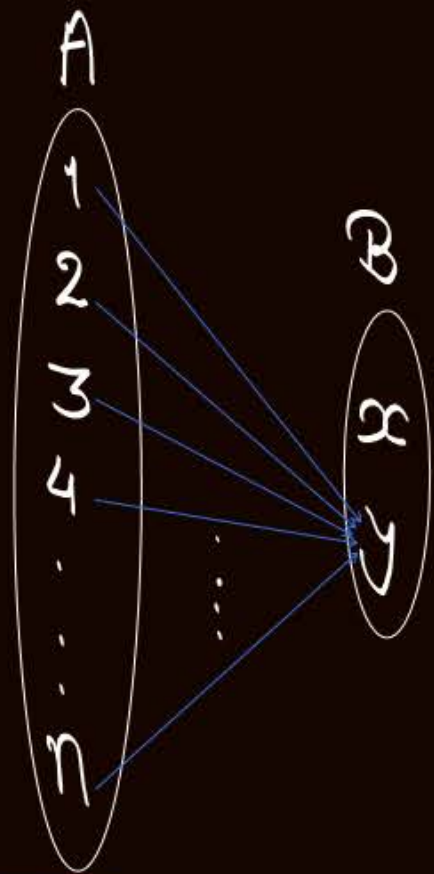
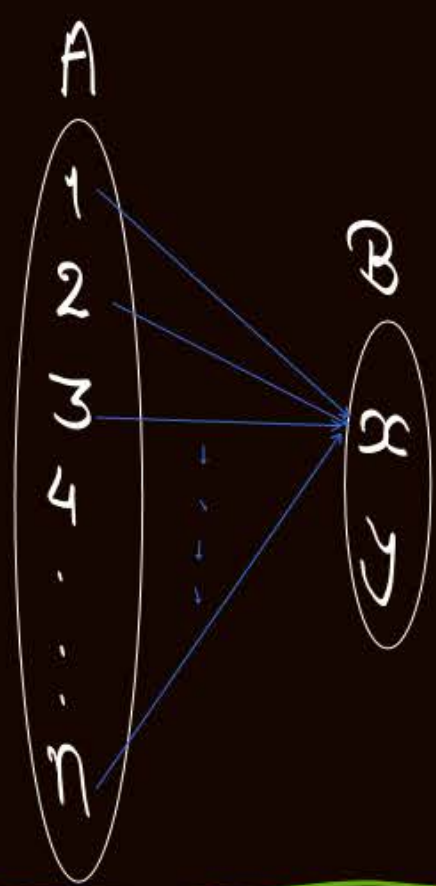
$$\text{Number of onto functions from set A to set B} = \text{Number of one-one functions from set A to set B}$$

Hence, if $|A| = |B| = n$, then

$$\text{No. of onto functions from set A to set B} = \text{No. of one-one functions from set A to set B} = n!$$

Q:- If $|A|=n$ and $|B|=2$ (s.t. $n \geq 2$)
 then How many onto functions are possible from A to B.

Number of Onto functions Possible from A to B = Total no. of functions Possible from A to B — No. of functions from A to B that are not onto

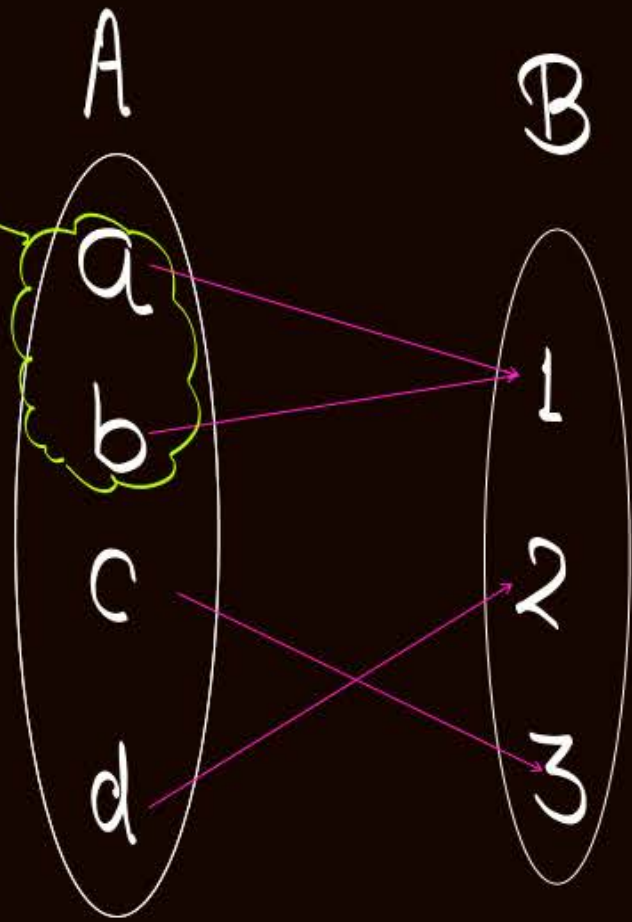


$$= 2^n - 2$$

These are the only two functions possible from Set A to Set B that are not Onto

Q:- If $|A|=4$ and $|B|=3$
then How many onto functions are possible from A to B.

Exactly two
elements of
Set A will
relate with
same element
of Set B.



↓
So Choose
two elements
in 4C_2 ways
& think of them
as a single element

Total no. of
onto functions
possible

$$\begin{aligned} &= {}^4C_2 * 3! \\ &= 6 * 6 \\ &= 36 \end{aligned}$$

Q:- If $|A|=n$ and $|B|=n-1$ i.e. $|A|=1+|B|$
then How many onto functions are possible from A to B .

$$\text{Total no. of} \\ \text{Onto functions} \\ \text{Possible} = {}^nC_2 * (n-1)!$$

Q: if $|A|=m$ & $|B|=n$ (such that $m \geq n$)
 then how many onto functions are possible from set A to set B.

$$\begin{aligned}
 \text{Total no. of} & \\
 \text{Onto functions} & \\
 \text{Possible from} & \\
 \text{Set A to set B} & \quad ||| \\
 &= n^m - nC_1(n-1)^m + nC_2(n-2)^m - nC_3(n-3)^m + nC_4(n-4)^m - \dots + (-1)^{n-1} nC_{(n-1)}(n-(n-1))^m \\
 &= nC_0(n-0)^m - nC_1(n-1)^m + nC_2(n-2)^m - nC_3(n-3)^m + \dots - \dots + (-1)^{n-1} nC_{n-1}(n-(n-1))^m
 \end{aligned}$$

$$= \sum_{i=0}^{(n-1)} (-1)^i \cdot nC_i \cdot (n-i)^m$$

		No. of onto functions from set A to set B
①	$ A =n$ & $ B =2$ ($n \geq 2$)	$2^n - 2$
②	$ A =n$ & $ B =n$	$n!$
③	$ A =n$ & $ B =n-1$ $ A = 1 + B $	$n \cdot 2^{n-2} \cdot (n-1)!$
④	$ A =m$ & $ B =n$ Where ($m \geq n$)	$\sum_{i=0}^{n-1} (-1)^i \cdot n \cdot C_i \cdot (n-i)^m$



2 mins Summary



- ✓ **Topic** Functions
- ✓ **Topic** Total number of functions
- ✓ **Topic** One-one and onto functions
- ✓ **Topic** Number of one-one functions
- ✓ **Topic** Number of onto functions

THANK - YOU